Global Seismic Series: Statistical Analysis of Spatiotemporal Clustering in M>=6.5 Earthquake Catalogs (2150 BCE – 2026 CE)

Yaroslav Marshalkin
Corresponding author: marshalkin@gmail.com
Telegram: @MRSHLKN | GitHub: github.com/marshalkin-ux/paleoseismic-clustering

Abstract.

We analyse a merged catalog of 4,267 unique M>=6.5 events (from 4,418 CSV rows; ~151 NOAA M<6.5 excluded from clustering; 2150 BCE–2026 CE) using the Baiesi–Paczuski metric eta with tectonic-path distance (Bird 2003). 47 global seismic series are identified (27 modern, 15 early, 5 historical candidates). Significance: permutation test (n=10,000, p<0.0001, z=-6.17); ETAS validation (FPR=0/100, seed=42); FDR (45/47 at q=0.05). Series detection is a statistical anomaly, not proof of causal triggering. Largest series: 1905–1910 (193 events, 43 regions); most extensive modern: S170 (46 events, 12 regions, Mmax=9.1).

Keywords: global seismicity; seismic series; earthquake clustering; tectonic distance; Baiesi–Paczuski metric; ETAS validation; false discovery rate; Monte Carlo; paleoseismology; remote triggering

1. INTRODUCTION

Large earthquakes do not occur as independent Poisson events. Following the 1992 Landers earthquake (M_w 7.3), Hill et al. [1993] documented remotely triggered seismicity at distances exceeding 1,000 km. Brodsky & Prejean [2006] showed that surface waves can initiate swarms in volcanic systems thousands of kilometres away. The 2004 Sumatra–Andaman earthquake (M_w 9.1) was followed by elevated activity in distant regions [Pollitz et al., 1998; Freed & Lin, 2001].

However, the systematic nature of such correlations remains debated. Michael [2011] found that M>=7 clustering in 1995–2011 is statistically indistinguishable from random fluctuations. Shearer & Stark [2012] reported no increase in global M>=7 and M>=8 rates after the 2004 Sumatra event. Kagan & Jackson [1999] confirmed elevated probability of paired events at short separation without resolving long-range links.

The ETAS model [Ogata, 1988] reproduces regional aftershock clustering but does not encode inter-plate correlations. The Baiesi–Paczuski [2004] metric and Zaliapin–Ben-Zion extensions [2008, 2013] provide objective cluster detection but typically use Euclidean distance, ignoring lithospheric connectivity.

Objective. We test the hypothesis that multi-regional seismic series exist in a four-millennium catalog of M>=6.5 earthquakes, using an adapted eta metric with tectonic distance along Bird (2003) plate boundaries. Novelty. To our knowledge, this is the first global application of nearest-neighbor clustering with tectonic-path distance across historical, early instrumental, and modern catalogs, combined with ETAS validation and FDR correction.

2. DATA AND METHODS

2.1 Catalog compilation

Three catalogs were merged: USGS ComCat (1900–2026, primary instrumental); ISC Bulletin (relocated hypocenters); and NOAA NGDC (historical and paleoseismic records from ~2150 BCE). Duplicates were merged using +/-30 days and <=50 km tolerance; source priority: ISC > USGS > NOAA. After deduplication, the catalog contains 4,267 unique M>=6.5 events (from 4,418 CSV rows; ~151 M<6.5 excluded from clustering). USGS ComCat: 2,088 raw (1973–2026) → 2,041 after merge/ISC. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041). quality_score is metadata, not an inclusion filter. Final catalog: 4,267 M>=6.5 events (4,418 CSV rows).

2.1.1 Catalog completeness

Maximum-curvature analysis yields M_c = 6.55. Maximum-likelihood b-value from 1,688 events above M_c:

b = 0.911 +/- 0.018 (n = 1,688 events)

2.2 Tectonic distance and connectivity metric

Tectonic distance r_{ij} is the shortest path between hypocenters along the Bird (2003) plate-boundary graph (20 key segments). Paths are computed with Dijkstra's algorithm (NetworkX).

If either hypocenter lies >500 km from the nearest boundary node, or no graph path exists, r_{ij} = 1.5 x r_{GC} (great-circle Haversine). Fallback audit (analyze_tectonic_fallback.py, fig07): 4987 pairs from 500 events; 98.0% GC fallback, 2.0% Dijkstra (4015 snap>500 km, 872 no path, 100 Dijkstra; 95 materially different; examples FE31 Japan, FE35 Philippines). Metric adds value for ~2% boundary-proximal pairs only. Tectonic diagnostic: median Deltalog10eta = +0.28.

eta_ij = t_ij * r_ij^1.6 * 10^(-b*mi)

b=1.0 per Baiesi & Paczuski (2004) for eta comparability; catalog b=0.911+/-0.018 in Monte Carlo null only.

2.3 Analysis pipeline

Raw catalogs → dedup → exclude M<6.5 (~151 NOAA) → GK declustering (primary) → eta NN forest → sliding windows → merge overlapping → series criteria → MC/ETAS/FDR. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041) — separate sensitivity check, not sequential filter.

2.4 Threshold eta_0, series detection, and ETAS parameters

Automatic eta_0 from nearest-neighbor eta distribution: KDE valley between bimodal log_{10}(eta) modes (Zaliapin & Ben-Zion 2013); default eta_0 = $10^{\text{median log10 eta}}$. Validated against Poisson permutation null (n = 10,000).

1. Declustering via Gardner–Knopoff [1974].
2. Nearest-neighbor forest: parent i* = argmin eta_{ij} subject to eta_{ij} < eta_0.
3. Global series: N >= 4; M >= 6.5; >= 3 Flinn–Engdahl regions.
4. Sliding windows (1, 2, 5 yr); overlapping groups merged.

ETAS parameters (not catalog-calibrated): Helmstetter & Sornette 2003 defaults (mu=0.008, K=0.08, alpha=1.0, c=0.005 d, p=1.1); not refit on 2041 events (results/etas_calibration_note.md). Optional MLE mu ~0.105 events/day vs default 0.008.

2.5 Statistical validation

Permutation test: n = 10,000, p < 0.0001, z = -6.17 (modern). ETAS validation: 100 catalogs (seed=42); FPR=0/100; p_ETAS=0.0000. Limitation: FPR=0/100 at seed=42 only; multi-seed future work. FDR (q=0.05): 45/47 significant. Declustering: GK 2,017/2,041; ZBZ 2,040/2,041.

3. RESULTS

3.1 Identified series

Full historical analysis yields 47 global seismic series: 27 modern (p < 0.0001), 15 early instrumental (p = 0.007; pre-1960 incompleteness caveat), 5 historical candidates (not significant, p = 0.46; 47 M>=6.5 events pre-1900). 142 cluster candidates before filtering.

ID	N	Regions	M_{max}	Period	Notes
1905–1910	193	43	8.8	1905–1910	Largest series; early instrumental

S047	53	5	8.0	1982–2024	Western Pacific subduction corridor
S170	46	12	9.1	2002–2023	Sunda belt; Sumatra 2004 (M 9.1)
S095	25	4	7.9	1989–2017	Western Pacific arc
S116	22	5	8.2	1993–2021	South Pacific multi-arc series

Table 1. Top five multi-regional series (>= 3 Flinn–Engdahl regions).

3.2 Spatial–temporal distribution

Elevated series activity occurs in 1952–1965 and 2002–2016 (post-Sumatra period). Spatially, clusters concentrate along the circum-Pacific belt (Kamchatka, Kuril Islands, Japan, Tonga, Indonesia). Tectonic diagnostic: median Deltalog10eta = +0.28 on random pairs (mostly 1.5x GC fallback).

4. DISCUSSION AND CONCLUSIONS

Statistical anomaly (established): series incompatible with Poisson and local-only ETAS nulls; FDR 45/47. Conclusion about eta links and p-values, not causality. Working hypotheses (not claims): viscoelastic mantle coupling (Pollitz 1998), dynamic triggering (Hill 1993; Brodsky 2006), shared tectonic loading (Freed & Lin 2001). S170 Delta CFS: Okada 1985, 79 receivers 2004–2016; Japan +0.008 kPa (n=63), Aleutians +0.000065 kPa (n=16); << 0.1 bar (fig06_cfs_s170.png). ETAS seed limitation: FPR=0/100 at seed=42 only. Tectonic audit: 98% GC fallback of 4987 pairs (fig07).

1. 4,267 M>=6.5 events (4,418 CSV records) contain 47 global series; 27 modern significant at $p < 0.0001$.
2. ETAS validation (mu=0.008, K=0.08, alpha=1.0, c=0.005 d, p=1.1, 500 km; FPR=0/100) and FDR (45/47) confirm non-randomness.
3. Largest series: 1905–1910 (193 events, 43 regions, M_{max}=8.8); most spatially extensive modern: S170 (46 events, 12 regions, 2002–2023, M_{max}=9.1).
4. Tectonic path: 2.0% Dijkstra / 98.0% GC fallback (4987 pairs, 500 events); value mainly for boundary-proximal pairs (~2%).
5. S170 Delta CFS: Japan +0.008 kPa (n=63), Aleutians +0.000065 kPa (n=16); promoting but << 0.1 bar.
6. Interpretive fork: (a) if eta links are real — hazard implications without claiming direct triggering; (b) if artifacts — FDR+ETAS remains reproducible null-test.

Future work: Delta CFS for S047, S095 with GCMT; full ETAS MLE; multi-seed; Zenodo.

DATA AND CODE AVAILABILITY

Data and code: github.com/marshalkin-ux/paleoseismic-clustering (docs/data_availability.md).

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